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Multi-modal temporal attention models for crop mapping from satellite time series

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Highlights

- PASTIS-R first open-access multimodal dataset with panoptic annotations.
- Thorough evaluation of different fusion schemes on three crop mapping tasks.
- Multimodality brings consistent improvement in performance and cloud robustness.

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Abstract

Optical and radar satellite time series are synergetic: optical images contain rich spectral information, while C-band radar captures useful geometrical information and is immune to cloud cover. Motivated by the recent success of temporal attention-based methods across multiple crop mapping tasks, we propose to investigate how these models can be adapted to operate on several modalities. We implement and evaluate multiple fusion schemes, including a novel approach and simple adjustments to the training procedure, significantly improving performance and efficiency with little added complexity. We show that most fusion schemes have advantages and drawbacks, making them relevant for specific settings. We then evaluate the benefit of multimodality across several tasks: parcel classification, pixel-based segmentation, and panoptic parcel segmentation. We show that by leveraging both optical and radar time series, multimodal temporal attention-based models can outmatch single-modality models in terms of performance and resilience to cloud cover. To conduct these experiments, we augment the PASTIS dataset (Garnot and Landrieu, 2021a) with spatially aligned radar image time series. The resulting dataset, PASTIS-R, constitutes the first large-scale, multimodal, and open-access satellite time series dataset with semantic and instance annotations. (Dataset available at: https://zenodo.org/record/5735646)

Keywords

Deep learning; Temporal attention; Multi-temporal fusion; Data fusion; SAR; Sentinel satellite

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Introduction

The multiplication of Earth Observation satellites with various sensors represents an opportunity to improve the performance of automated analysis of remote sensing data for tasks such as crop mapping. Indeed, different modalities capture information of different natures and distinct spatial and temporal resolutions and have varying resilience to atmospheric conditions. Machine learning models can leverage these complementary characteristics to learn richer and more robust representations. In particular, C-band radar and optical images possess well-known synergies for automated crop mapping (Van Tricht et al., 2018, Steinhausen et al., 2018, Campos-Taberner et al., 2019), the driving application of this paper. More specifically, multispectral time series contain highly relevant information for monitoring the evolution of plant phenology (Vrieling et al., 2018, Segarra et al., 2020). For example, the study of red and infrared reflectances helps monitoring photosynthetic activity (Tucker, 1979). However, passive optical sensors are highly susceptible to cloud cover and atmospheric distortion (Sudmanns et al., 2020). Conversely, due to the influence of extrinsic factors such as humidity and terrain, it is harder to extract discriminative information from radar images for crop mapping. On the other hand, the high revisit frequency and imperviousness to cloud cover make them uniquely well-suited for monitoring the rapid-changing biological processes of agricultural parcels (McNairn et al., 2014).

Automated crop mapping is necessary for various applications carrying crucial economic and ecological stakes, such as environmental monitoring, subsidy allocation, and food price prediction. For example, the Common Agricultural Policy is responsible for the allocation of over 57 billion euros each year of agricultural subsidies in the European Union (E. Commission, 2016). This application is at the center of an effort towards algorithmic solutions for machine learning-based crop monitoring (Koetz et al., 2019). This endeavor is aided by the accessibility of high-quality satellite data worldwide (Drusch et al., 2021) and individual parcel annotations for some European countries such as France (EtaLab, 2017), which is particularly conducive to training large-scale deep networks.

In the context of crop type mapping, the fusion of optical and radar time series has been extensively explored with traditional machine learning methods (Van Tricht et al., 2018, Steinhausen et al., 2018, He and Yokoya, 2018, Campos-Taberner et al., 2019, Orynbaikyzy et al., 2020, Giordano et al., 2020), and more recently recurrent neural networks (Ienco et al., 2019). However, despite the significant performance gain offered by methods based on temporal attention (Rußwurm and Körner, 2020, Garnot et al., 2020, Kondmann et al., 2021, Garnot and Landrieu, 2020), these approaches are mostly restricted to the analysis of optical Satellite Image Time Series (SITS). Recently, Ofori-Ampofo et al. (2021) proposed a first exploration of the benefit of fusion strategies for parcel-based crop type classification from Sentinel-1 and Sentinel-2 time series with attention-based methods. The present paper extends their analysis to a broader set of crop mapping tasks, including semantic and panoptic segmentation(Kirillov et al., 2019, Garnot and Landrieu, 2021c) of agricultural parcels, as summarised on Fig. 1. We also study the performance benefit of standard enhancements such as auxiliary supervision and temporal dropout.

To train and evaluate our models, we augment the open-access PASTIS optical time series dataset (Garnot and Landrieu, 2021a) with corresponding Sentinel-1 radar acquisitions for each 2433 time series for a total of $339,174$ radar images. We demonstrate that the right choice of fusion scheme can lead to improvement across the board for all tasks and increased robustness to varying cloud cover. The main contributions of this paper are as follows:

- We present a complete reformulation of fusion strategies in the context of temporal attention-based SITS encoders, as well as standard model enhancements.
- We present PASTIS-R, the first large-scale, multimodal, open-access SITS dataset with panoptic annotations (https://zenodo.org/record/5735646 ↗).

- We evaluate our fusion schemes and enhancements on parcel classification, semantic and panoptic segmentation, defining a new state-of-the-art for all tasks.
- We show that combining optical and radar imagery grants significant improvement in terms of robustness to varying cloud cover.

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Related work

In the following paragraphs, we review the recent literature on fusion approaches for the multi-temporal fusion of SITS. In particular, we detail commonly implemented fusion strategies.

Traditional Approaches for Multimodal SITS. Multiple traditional machine learning approaches such as random forest or support vector machines have been adapted to handle information from optical and radar images. As highlighted by the review of Joshi et al. (2016), the joint processing of both modalities can …

Methods

We consider a set of *M* image time series $\{X^m\}_{m=1}^M$ corresponding to *M* distinct modalities for a single geo-referenced patch containing one or several agricultural parcels. We assume that all modalities are resampled to the same resolution for simplicity's sake. Each time sequence X^m can be expressed as a tensor of size $T^m \times C^m \times H \times W$ with T^m the number of available temporal acquisitions for modality m, C^m the feature size for each pixel for the modality *m*, and $H \times W$ the spatial extent of the patch. …

Experiments

We present in this section our numerical experiments to assess the benefit of multimodality for crop mapping with temporal attention-based networks. We evaluate several modality-fusion schemes and several mapping tasks. We also introduce a new large-scale open-access and multimodal dataset with annotations fit for all tasks. …

Discussion

In this section, we discuss the relevance of the different modality fusion strategies, with a focus on Sentinel-1 & 2 data for crop mapping. Our experiments showed that combining optical and radar imagery allowed for an increase in performance for all tasks considered (Table 1, Table 2, Table 3) as well as robustness to cloud cover (Fig. 7). …

Conclusion

This article formulated and explored different schemes to design fusion architectures using temporal attention for predicting agricultural crop type maps from radar and optical satellite time series.

Across the three tasks of parcel-based classification, semantic segmentation, and panoptic segmentation, we experimentally confirmed that the multispectral information of Sentinel-2 proves more discriminative than the SAR signal of Sentinel-1. Yet, for all three tasks, leveraging both modalities led …

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. …

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References (63)

D. Tarchi *et al.*
Landslide monitoring by using ground-based sar interferometry: an example of application to the tessina landslide in italy
Eng. Geol. (2003)

S. Abramov *et al.*
Speckle reducing for sentinel-1 sar data
IGARSS (2017)

P. Benedetti *et al.*
M3Fusion: A deep learning architecture for multiscale multimodal multitemporal satellite data fusion
JSTARS (2018)

Campos-Taberner, M., García-Haro, F.J., Martínez, B., Sánchez-Ruíz, S., Gilabert, M.A., 2019. A Copernicus Sentinel-1…

E. Christophe *et al.*
Orfeo toolbox: a complete solution for mapping from high resolution satellite images
Int. Arch. Photogramm. Remote Sens. Spatial Inform. Sci. (2008)

S.R. Cloude *et al.*
A review of target decomposition theorems in radar polarimetry
Trans. Geosci. Remote Sens. (1996)

Drusch, M., Del Bello, U., Carlier, S., Colin, O., Fernandez, V., Gascon, F., Hoersch, B., Isola, C., Laberinti, P.,…

E. Commission, 2016. The common agricultural policy at a glance.…

EtaLab, 2017. Registre parcellaire graphique (rpg): contours des parcelles et ı̂lots culturaux et leur groupe de…

Garioud, A., Valero, S., Giordano, S., Mallet, C., 2020. On the joint exploitation of optical and SAR satellite imagery…

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